

5-Mark Questions & Answers — Data Structures Module 4: Trees

Q1. Explain Binary Search Tree (BST) operations: Search, Insert, and Delete (all 3 cases) with C algorithms.

- **Search / Insert:** Follow binary decision (Key < Node: go left, Key > Node: go right).
- **Delete Cases:**
 - Case 1 (0 Child): Free node and set pointer to NULL.
 - Case 2 (1 Child): Bypass node with its single child.
 - Case 3 (2 Children): Replace with **Inorder Successor** (smallest in right subtree) and recursively delete successor.

Q2. Construct a Huffman Tree and compute binary codes for the following character frequencies:

A: 45, B: 13, C: 12, D: 16, E: 9, F: 5

Min-Heap merges:

1. Merge F(5) + E(9) = Node(14)
2. Merge C(12) + B(13) = Node(25)
3. Merge D(16) + Node(14) = Node(30)
4. Merge Node(25) + Node(30) = Node(55)
5. Merge A(45) + Node(55) = Root(100)

- Codes: A=0, C=100, B=101, D=110, F=1110, E=1111 .

Q3. Explain B-Tree properties and compare B-Tree with B+ Tree in database indexing.

- Properties of order m : Max m children, min $\lceil m/2 \rceil$ children, all leaves at same level.
- B+ Tree advantages: Higher fanout, faster sequential range queries via doubly linked leaf nodes.